Su Mo Tu We Th Fr Sa

ASSIGNMENT - 1

Date :

String Handling, Indexing & Ranging :

① s = "Python"

print (s[0], s[-1], s[6])

Ans : Error. s[0] → P s[-1] → n

Because the s[6] does not exist, it is out of range.

② Positive Indexing :-

It access a character using its location number.

It is traversing

Eg :- name = 'Priya'

name [0]	name [1]
'P'	'r'

Negative Indexing :

It is back traversing.

Eg :- name = 'Dharshini'

name [-1]	name [-2]
'i'	'n'

s = "Hello" s[-2] = 'l'

Su Mo Tu We Th Fr Sa

Date :

③ It shows error when the index that is out of range in string.

In slicing it gives available character.

```
s = "code"
print(s[10])   'error'
print(s[1:10])  'ode'
```

④ s = "mississippi"
0 1 2 3 4 5 6

Index of first 's' = 2

last 's' = 6

⑤ s = "abcdef"
-6 -5 -4 -3 -2 -1

```
print(s[-1], s[-3], s[-6])
        (f, d, a)
```

In negative indexing it counts from backward of the string.

Su Mo Tu We Th Fr Sa

Date :

⑥. String slicing in python is getting a particular position of string.

s[start:stop:step].

Start → The index where the slicing starts.

Stop → The index where the slicing stops.

Step → Intermediate value which is extracted.

⑦. s = "0 1 2 3 4 5 6 7 8 9"
"Hello World"

Print (s[0:5]) → Hello

Print (s[5:]) → World

Print (s[:5]) → Hello

Print (s[:]) → HelloWorld

⑧ In positive step the start should be smaller than stop so it returns the blank space.

Su Mo Tu We Th Fr Sa

Date :

(9). By using negative index we can extract the last 3 char of a string using slicing.

(10). s = "abcdefgh"

print(s[::1]) → abcdefgh

print(s[::2]) → aceg

print(s[1::2]) → bdfh

(11). s = "Python"

print(s[::-1])

output → nohtyP

The step value is (-1).

(12). s = "abcdefgh"

Print(s[::-1]) → hgfedcba

print(s[::-2]) → hfdb

print(s[6::-1]) → gfedc

→ The string is scanned from backward. (right to left).

Su Mo Tu We Th Fr Sa

Date :

(13). s = "a1b2c3d4e5"

Print(s[1:10:3])

output : "12345".

(14). s = "Python"

print(s[5:0:-1]) -> nohty

print(s[5::-1]) -> nohtpy

print(s[:0:-1]) -> nohty

start -> end of the string.

Stop -> beginning of the string.

(15). s = "hello"

print(s[100::-1]) -> olleh.

print(s[::-100]) -> o

It automatically counting them to the string boundaries.

(16). s = "abcdefghij"

print(s[2:-2])

'cdefgh'

Su Mo Tu We Th Fr Sa

Date :

(17). s = "racecar"

print(s[::-1])

'racecar'.

It is plaindrome.

'True'.

(18). s = "Hello"

Print(s[1:100]) → ello

Print(s[-100:3]) → hel

print(s[100:200]) → ___ (blank space)

python does not raise an error because slicing handles out of range.

(19). s = "Hello world"

print(s[:6] + "python")

Hello Python.

(20). s = "abcdefghij"

my_slice = slice(1,8,2)

Print(s[my_slice])

output :- bdfh.